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Roll No

CE-703(C)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Integrated Waste Management

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the adverse health and environmental impacts due to improper handling of solid waste. 7
b) Estimate the Energy content, Energy content on dry basis and Energy content on an ash-free dry basis for a waste sample with the given composition, and choose the correct range: 7
(Assume moisture content as 20% as Ash content as 5%)

Component	% by mass	Energy, kJ/kg
Food Waste	10	4,650
Paper	45	16,750
Cardboard	14	16,300
Plastic	6	32,600
Garden trimming	15	6,500
Wood	5	18,600
Tin cans	5	700

2. a) Write seven benefits of recycling of waste? 7
b) Explain the necessity of source reduction in solid waste management. 7

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3. a) Solid waste from a commercial area are to be collected using a stationary container collection system having 6 m^3 containers. Determine the appropriate truck capacity for the following conditions: 7
- i) Container size = 6 m^3
 - ii) Container utilization factor = 0.8
 - iii) Average number of containers at each location = 2
 - iv) Collection vehicle compaction ratio = 3.0
 - v) Container unloading time = 0.1 h/container
 - vi) Average drive time between container locations = 0.1 h
 - vii) One-way haul distance = 30 km
 - viii) Speed limit = 88 km/h (55 mi/h)
 - ix) Time from garage to first container location = 0.3 h
 - x) Time from last container location to garage = 0.2 h
 - xi) Number of trips to disposal sites per day = 2
 - xii) Length of workday = 8 h
 - xiii) Use Length of workday as 8 h and off-route factor as 0.2
 - xiv) $a = 0.016 \text{ h/trip}$, $s = 0.1 \text{ h/trip}$
 - xv) Empirical haul constant $b = 0.011 \text{ h/km}$
- b) Explain Hauled container System with neat and detailed sketch. 7
4. a) Discuss human health risks due to composting and possible preventive measures. 7
- b) How does incineration help in the management of solid waste? Describe the incineration technologies and air emissions and its control in detail. 7
5. a) Explain the various phases of municipal solid waste decomposition in a closed landfill cell. 7
- b) Identify the adverse effects of a landfill leachate and list appropriate control measures. 7

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6. a) Write a note on waste management under Swachh Bharat Mission and Smart City Program. (400 words) 7
b) Describe the six ways of reuse of waste in industry and its limitations. 7
7. a) What do you mean by solid waste management hierarchy? Draw neat sketch and explain. 7
b) Describe any two-leachate control method with a neat sketch. 7
8. Following data is given for a town in Madhya Pradesh, find out the required area of landfill for next three years. 14
 - a) Population = 50,000
 - b) Population growth rate = 8% per year
 - c) Per capita generation rate = 0.40 kg/day/person
 - d) Per capita production growth rate = 2 % per year
 - e) $P_n = P_o (1+rp)^n$
 - f) $GR_n = GR_o (1+rw)^n$
 - g) Density of stabilized soil mass = 600 kg/m³
 - h) Depth of landfill = 3.1m (7m)
 - i) Cover of soil = Zero (0)
